

Classifier Fairness

Fraida Fund

In addition to evaluating the **error** of a classifier, we are also often concerned with the **fairness** of a classifier.

Suppose samples come from two groups: a and b . What does it mean for a classifier to treat both groups *fairly*? There are a number of different types of fairness, and like the error metrics described previously, we are often stuck in a situation where we must sacrifice on one fairness measure to improve another.

Fairness metrics

This section lists some metrics related to fairness. You won't have to memorize these, but you should understand them, and given the definition of any of these metrics, you should be able say whether or not it is satisfied in a particular scenario.

- **Fairness through unawareness** is not a measure of fairness, but describes a situation in which features related to group membership are not used in classification. (This doesn't necessarily mean that the classifier produces fair outcomes! For example, if the classifier is trained on data that reflects an underlying bias in society, the classifier will be biased even if it is not trained on features related to group membership.)
- **Causal discrimination** says that two samples that are identical w.r.t all features except group membership, should have same classification.
- **Group fairness** (also called *statistical parity*) says that for groups a and b , the classifier should have equal probability of positive classification:

$$P(\hat{y} = 1 | G = a) = P(\hat{y} = 1 | G = b)$$

- **Conditional statistical parity** is a related metric, but now we are also controlling for factor F :

$$P(\hat{y} = 1 | G = a, F = f) = P(\hat{y} = 1 | G = b, F = f)$$

- **Balance for positive/negative class** is similar to *group fairness*, but it is for classifiers that produce soft output. It applies to every probability S produced by the classifier. This says that the expected value of probability assigned by the classifier should be the same for both groups -
- For **positive class balance**

$$\mathbb{E}[S | y = 1, G = a] = \mathbb{E}[S | y = 1, G = b]$$

- For **negative class balance**

$$\mathbb{E}[S | y = 0, G = a] = \mathbb{E}[S | y = 0, G = b]$$

- **Predictive parity** (also called *outcome test*) says that the groups should have equal PPV, i.e. the prediction should carry similar meaning (w.r.t. probability of positive outcome) for both groups:

$$P(y = 1 | \hat{y} = 1, G = a) = P(y = 1 | \hat{y} = 1, G = b)$$

- Predictive parity also implies equal FDR:

$$P(y = 0|\hat{y} = 1, G = a) = P(y = 0|\hat{y} = 1, G = b)$$

- **Calibration** (also called *test fairness*, *matching conditional frequencies*) is similar to *predictive parity*, but it is for classifiers that produce soft output. It applies to every probability S produced by the classifier:

$$P(y = 1|S = s, G = a) = P(y = 1|S = s, G = b)$$

- **Well-calibration** extends this definition to add that the probability of positive outcome should actually be s :

$$P(y = 1|S = s, G = a) = P(y = 1|S = s, G = b) = s$$

- **False positive error rate balance** (also called *predictive equality*) says that groups should have equal FPR:

$$P(\hat{y} = 1|y = 0, G = a) = P(\hat{y} = 1|y = 0, G = b)$$

- False positive error rate balance also implies equal TNR:

$$P(\hat{y} = 0|y = 0, G = a) = P(\hat{y} = 0|y = 0, G = b)$$

- **False negative error rate balance** (also called *equal opportunity*) says that groups should have equal FNR. Equal opportunity compares outcomes among actual positives; group fairness compares the positive prediction rate over the whole group. Equal prevalence alone does not make these criteria equivalent.

$$P(\hat{y} = 0|y = 1, G = a) = P(\hat{y} = 0|y = 1, G = b)$$

- False negative error rate balance also implies equal TPR:

$$P(\hat{y} = 1|y = 1, G = a) = P(\hat{y} = 1|y = 1, G = b)$$

- **Equalized odds** (also called *disparate mistreatment*) says that both groups should have equal TPR and FPR:

$$P(\hat{y} = 0|y = i, G = a) = P(\hat{y} = 0|y = i, G = b), i \in 0, 1$$

- With different prevalence, equalized odds and predictive parity conflict when the common TPR and FPR are both positive. The derivation below covers exceptions, including zero false positives.
- **Conditional use accuracy equality** says that the groups have equal PPV and NPV:

$$P(y = 1|\hat{y} = 1, G = a) = P(y = 1|\hat{y} = 1, G = b)$$

$$P(y = 0|\hat{y} = 0, G = a) = P(y = 0|\hat{y} = 0, G = b)$$

- **Overall accuracy equality** says that the groups have equal overall accuracy

$$P(\hat{y} = y|G = a) = P(\hat{y} = y|G = b)$$

- **Treatment equality** says that the groups have equal ratio of FN to FP, $\frac{FN}{FP}$

Impossibility result

The fairness metrics above differ in which population they condition on:

- **Equalized odds** compares predictions among samples with the same *true label*, $P(\hat{y} \mid y, G)$.
- **Predictive parity** compares true labels among samples with the same *prediction*, $P(y \mid \hat{y}, G)$.
- **Calibration** compares true labels among samples with the same *score*, $P(y \mid S, G)$.
- **Statistical parity** compares the positive prediction rate over the *whole group*, $P(\hat{y} = 1 \mid G)$.

In other words: the denominator is different in each case.

This is important when we consider groups with different prevalence (the rate at which the positive condition appears). Prevalence determines how a group splits into actual positives and actual negatives, and so how many of each end up among the predicted positives (or among the samples with a given score).

Suppose two groups have the same error rates among actual positives and among actual negatives, but one group has more actual positives. Then a larger fraction of that group's predicted positives will be true positives, so a criterion computed over predicted positives can't be equal too. When the groups have different prevalence, equalizing a metric over one population generally forces a metric over a different population to be unequal.

We can make this precise for a classifier that produces soft output. Let $S \in [0, 1]$ be the classifier score, and let the prevalence in each group be

$$p_a = P(y = 1 \mid G = a), \quad p_b = P(y = 1 \mid G = b),$$

with $0 < p_a, p_b < 1$ (both classes occur in both groups) and $p_a \neq p_b$. Suppose we want:

- well-calibration within each group,
- positive class balance: the mean score among actual positives is the same value $\bar{S}_1$ in both groups,
- negative class balance: the mean score among actual negatives is the same value $\bar{S}_0$ in both groups.

Step 1: well-calibration fixes the mean score. If scores are well-calibrated, then $\mathbb{E}[y \mid S, G = g] = S$, and by the law of total expectation

$$\mathbb{E}[S \mid G = g] = \mathbb{E}[y \mid G = g] = p_g.$$

For example, suppose five people receive score 0.2 and five receive score 0.8. If the scores are well-calibrated, there are $1 + 4 = 5$ positives, so the prevalence and the mean score are both 0.5.

Step 2: split the mean score by true class. The group mean score is a weighted average of the mean score among positives and among negatives:

$$\mathbb{E}[S \mid G = g] = p_g \mathbb{E}[S \mid y = 1, G = g] + (1 - p_g) \mathbb{E}[S \mid y = 0, G = g].$$

Substituting Step 1 and the two balance requirements gives one equation per group:

$$p_a \bar{S}_1 + (1 - p_a) \bar{S}_0 = p_a,$$

$$p_b \bar{S}_1 + (1 - p_b) \bar{S}_0 = p_b.$$

Step 3: solve. Subtracting the second equation from the first,

$$(p_a - p_b)(\bar{S}_1 - \bar{S}_0) = p_a - p_b.$$

Since $p_a \neq p_b$,

$$\bar{S}_1 - \bar{S}_0 = 1$$

Both means lie in $[0, 1]$, so this is only possible if $\bar{S}_1 = 1$ and $\bar{S}_0 = 0$, i.e. if we have **perfect prediction**.

So, with different prevalence, any classifier that makes errors or assigns uncertain scores *must* give up at least one of the three requirements.

To see this with numbers, suppose we satisfy two of the three requirements and check what happens to the third.

We'll imagine that group has 100 people; group a has prevalence $p_a = 0.2$ and group b has prevalence $p_b = 0.5$.

- **Well-calibration:** by Step 1, the mean score in each group must equal its prevalence. So the scores in group a add up to $100 \times 0.2 = 20$, and the scores in group b add up to $100 \times 0.5 = 50$.
- **Positive class balance:** this requires actual positives to have the same mean score in both groups, but it doesn't say what that value is. We'll use $\bar{S}_1 = 0.8$ as an example; any common value below 1 leads to the same conclusion. In group a , the 20 positives contribute $20 \times 0.8 = 16$ to the total. In group b , the 50 positives contribute $50 \times 0.8 = 40$.

Whatever is left of each group's total must come from its actual negatives:

Group	Positives	Negatives	Total score	From positives	Left for negatives	Mean score of negatives
a	20	80	20	16	4	$4/80 = 0.05$
b	50	50	50	40	10	$10/50 = 0.20$

Negative class balance fails: actual negatives in group b get higher scores, on average, than actual negatives in group a . The only way to make both columns match is $\bar{S}_1 = 1$, which leaves 0 for the negatives in both groups (perfect prediction).

(This argument follows [Kleinberg, Mullainathan, and Raghavan, *Inherent Trade-Offs in the Fair Determination of Risk Scores*](#).)

We can show a similar result for a hard decision classifier as well:

Suppose equalized odds holds, i.e. both groups have the same TPR and FPR. Write the common TPR as r and the common FPR as f . Within group g , Bayes' rule gives

$$PPV_g = P(y = 1 \mid \hat{y} = 1, G = g) = \frac{rp_g}{rp_g + f(1 - p_g)}.$$

For $r > 0$ and $f > 0$, PPV increases with prevalence. If the two groups have different prevalence, equalized odds then rules out predictive parity.

For example, consider two groups of 1000 people each, with common $TPR = 0.8$ and $FPR = 0.1$:

Group	Prevalence	Actual positive	Actual negative	TP	FN	FP	TN	PPV
a	0.2	200	800	160	40	80	720	$160/240 \approx 0.67$
b	0.5	500	500	400	100	50	450	$400/450 \approx 0.89$

The classifier has the same error rates in both groups, but one-third of positive predictions in group a are false alarms, compared with one-ninth in group b .

Suppose a higher threshold for group a changes its TPR to 0.64 and FPR to 0.02, while group b keeps its threshold:

Group	TP	FN	FP	TN	PPV	TPR	FPR
a	128	72	16	784	$128/144 \approx 0.89$	0.64	0.02
b	400	100	50	450	$400/450 \approx 0.89$	0.80	0.10

We now have predictive parity but unequal TPR and FPR. Positive members of group a have a lower chance of receiving a positive prediction. Negative members of group b have a higher chance of receiving a false positive prediction. Whether those outcomes confer a benefit or impose a penalty depends on the application.

The exceptions require care. If $f = 0$ and $r > 0$, PPV is 1 in both groups, even if the classifier misses positives. If $r = 0$ and $f > 0$, PPV is 0 in both groups. If the classifier never predicts positive, PPV has a zero denominator and is undefined. Thus perfect prediction is sufficient for compatibility, but it is not the only exception for these hard decision criteria. See also [Chouldechova, Fair prediction with disparate impact](#).

Under equalized odds, the positive prediction rate in group g is

$$P(\hat{y} = 1 \mid G = g) = rp_g + f(1 - p_g).$$

The difference between groups is $(r - f)(p_a - p_b)$. With different prevalence, statistical parity and equalized odds can both hold only if $r = f$. At that operating point, the classifier predicts positive at the same rate among positives and negatives. In the first numerical example, 24% of group a and 45% of group b receive positive predictions despite equalized odds.

Disease screening activity

In [Measuring Fairness](#), a model screens children and adults for a disease that is more common among children. Try one threshold for both groups, then separate thresholds. Compare sensitivity (TPR), specificity (TNR), and precision (PPV). Can you make all three equal across groups?